

OZ — The Ozone Story

OCR B (Salters) H433 storyline **OZ**. Ozone as a sunscreen: radiation and matter, rates of reaction and catalysis, radical reactions, intermolecular bonding, and the haloalkanes of CFCs and HFCs with their nucleophilic substitution.

1. Rates of Reaction and Collision Theory

Collision Theory

Collision theory: For a reaction to occur, particles must collide with sufficient energy (equal to or greater than the activation energy, E_a) and with the correct orientation.

Activation energy (E_a): The minimum energy that colliding particles must have in order to react. It represents the energy barrier that must be overcome for bonds to break and new bonds to form.

Factors Affecting Rate

Factor	Effect	Explanation
Concentration (or pressure for gases)	Increases rate	More particles per unit volume → more frequent collisions → more successful collisions per unit time
Temperature	Increases rate	Particles have greater kinetic energy → more frequent collisions AND a greater proportion of particles exceed E_a (the latter is the dominant effect)
Surface area	Increases rate	More surface exposed to reactant particles → more frequent collisions

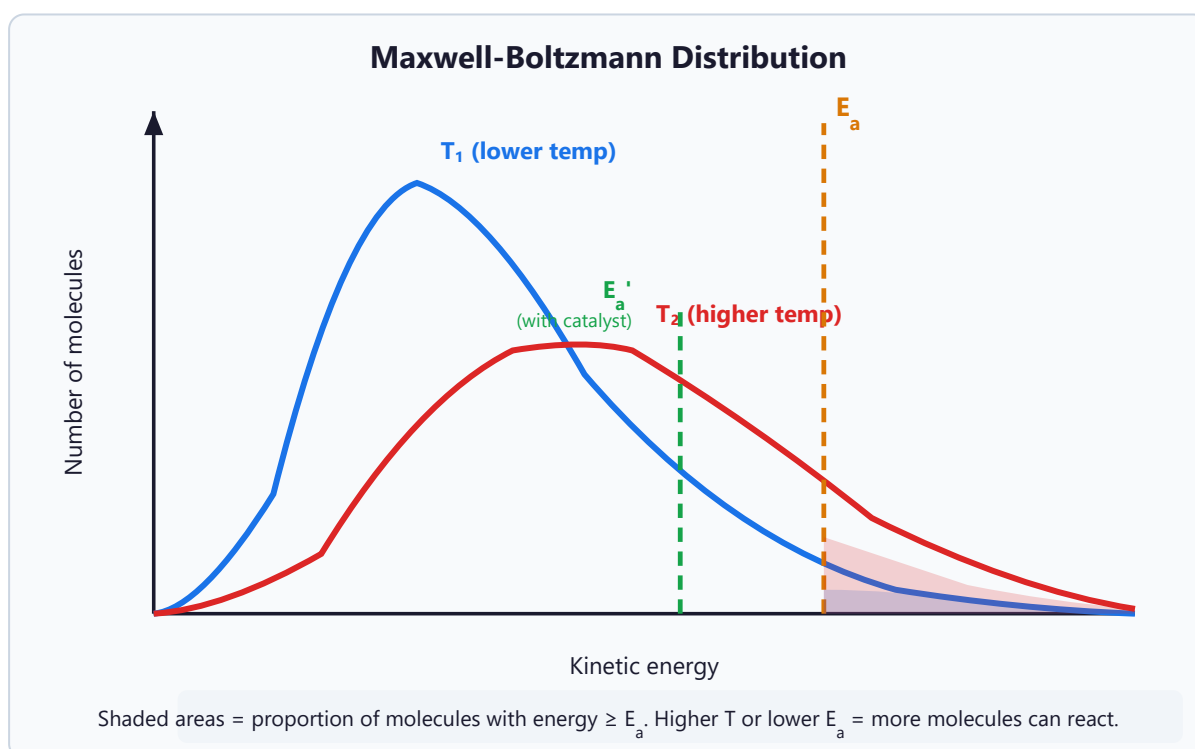
Catalyst	Increases rate	Provides an alternative reaction pathway with a lower activation energy
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The Boltzmann Distribution

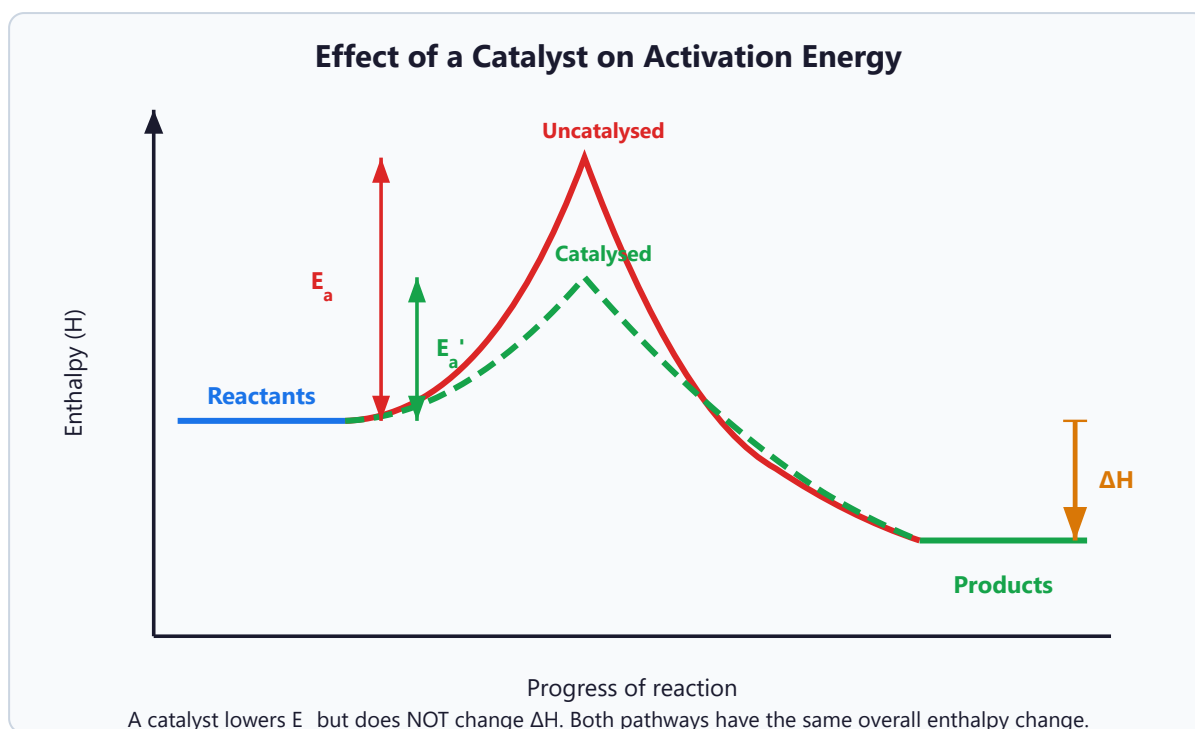
The Boltzmann (Maxwell-Boltzmann) distribution shows the distribution of molecular kinetic energies in a sample of gas at a given temperature. Key features:

- No molecules have zero energy.
- The distribution is asymmetric with a long tail to the right (some molecules have very high energies).
- The peak (most probable energy) shifts to the right and becomes lower and broader at higher temperatures.
- The area under the curve is constant (total number of molecules is unchanged).
- The shaded area to the right of E_a represents the proportion of molecules with enough energy to react.

Increasing temperature shifts the distribution to the right and significantly increases the proportion of molecules exceeding E_a . A catalyst lowers E_a , so a greater proportion of molecules exceed the new, lower E_a without changing the distribution itself.



Catalysts



Catalyst: A substance that increases the rate of a reaction by providing an alternative pathway with a lower activation energy, without being consumed in the overall reaction.

Homogeneous catalyst: A catalyst in the same phase as the reactants (e.g. $\text{H}^+(\text{aq})$ catalysing an esterification in aqueous solution).

Heterogeneous catalyst: A catalyst in a different phase from the reactants (e.g. iron in the Haber process, vanadium(V) oxide in the Contact process). Reactions occur on the surface of the catalyst, so a greater surface area increases effectiveness.

2. Halogenoalkanes

Halogenoalkane: An organic compound in which one or more hydrogen atoms of an alkane have been replaced by halogen atoms (F, Cl, Br, I). The C–X bond is polar because halogens are more electronegative than carbon, making the carbon atom susceptible to attack by nucleophiles.

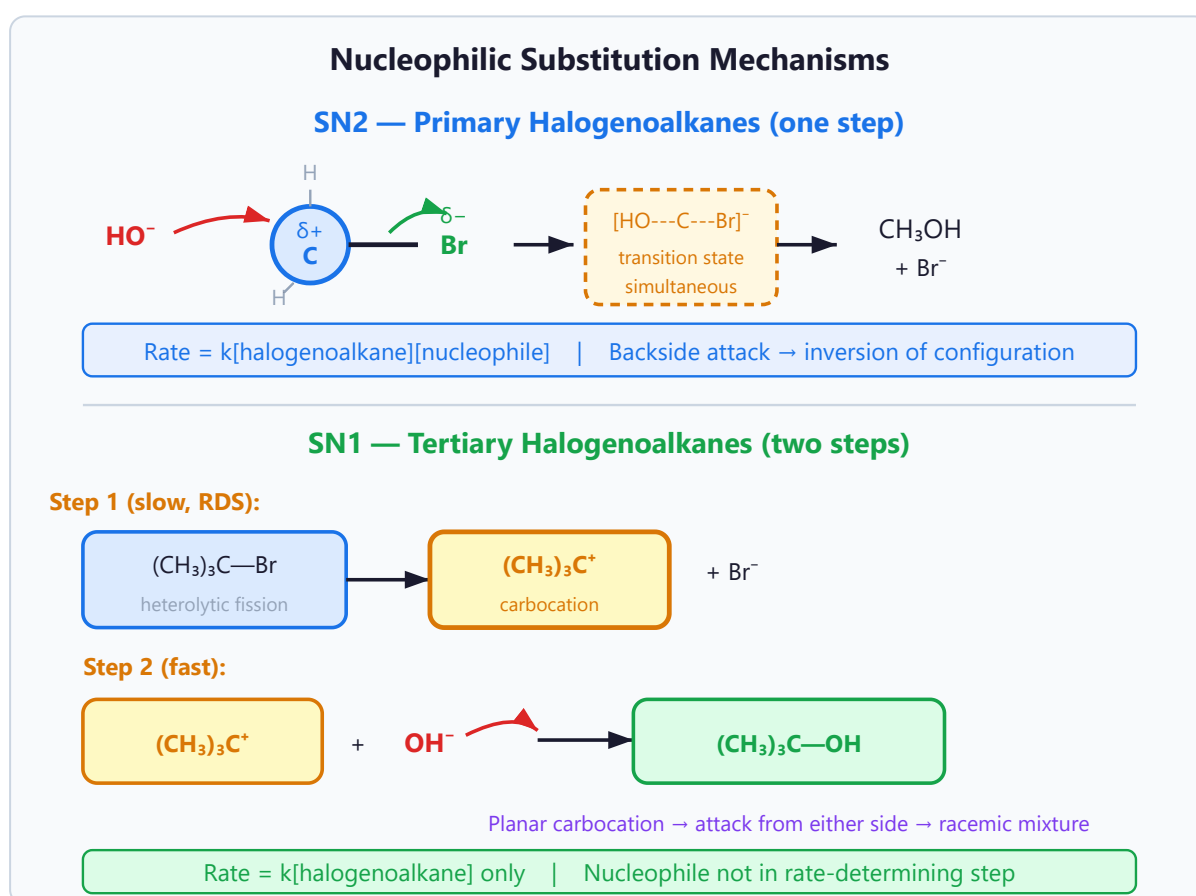
Nucleophilic Substitution

Nucleophile: An electron-pair donor; a species with a lone pair that it can donate to an electron-deficient atom. Examples: OH^- , CN^- , NH_3 , H_2O .

SN2 mechanism (primary halogenoalkanes): A one-step mechanism. The nucleophile attacks the δ^+ carbon from the opposite side to the halogen (backside attack). Bond making and bond breaking happen simultaneously in a single transition state. The rate equation is: $\text{rate} = k[\text{halogenoalkane}][\text{nucleophile}]$. SN2 produces inversion of configuration at the chiral centre.

SN1 mechanism (tertiary halogenoalkanes): A two-step mechanism. Step 1: the C–X bond breaks heterolytically to form a carbocation and a halide ion (slow, rate-determining step). Step 2: the nucleophile rapidly attacks the planar carbocation from either side. The rate equation is: $\text{rate} = k[\text{halogenoalkane}]$. SN1 produces a racemic mixture if the carbon was a chiral centre, because the planar carbocation can be attacked equally from both sides.

Secondary halogenoalkanes undergo a mixture of SN1 and SN2 mechanisms.

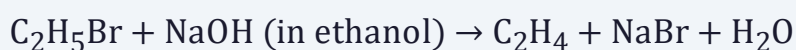


Reactions of Halogenoalkanes

Nucleophile	Conditions	Product
NaOH(aq) / OH ⁻	Warm aqueous solution	Alcohol
KCN / CN ⁻	Warm ethanol/water	Nitrile (extends carbon chain by 1)
NH ₃	Excess, warm, sealed tube	Amine

Elimination

When a halogenoalkane is heated with a strong base in ethanolic solution (e.g. NaOH in ethanol), elimination occurs instead of substitution. A hydrogen atom and the halogen are removed from adjacent carbons, forming an alkene and water:



Exam Tip: The choice between substitution and elimination depends on the conditions. Aqueous NaOH favours substitution (nucleophilic); ethanolic NaOH favours elimination (OH⁻ acts as a base). Tertiary halogenoalkanes favour elimination and S_N1; primary halogenoalkanes favour S_N2.

Worked Example — Percentage Yield:

A student reacted 5.00 g of butan-1-ol with excess ethanoic acid. The theoretical yield of the ester (butyl ethanoate, M_r = 116.2) is 7.85 g. The actual yield was 4.50 g.

$$\text{Percentage yield} = (4.50 / 7.85) \times 100 = \mathbf{57.3\%}$$

The yield is below 100% due to: the reversible nature of esterification, loss during transfer and separation, incomplete reaction, and evaporation of volatile ester.

Reactivity of Halogenoalkanes

The rate of hydrolysis (reaction with aqueous silver nitrate to test for the halide ion) decreases in the order: R-I > R-Br > R-Cl. This is because the C-I bond is the weakest (lowest bond enthalpy) and easiest to

break, despite being the least polar. Bond strength is the dominant factor, not polarity.

3. Organic Mechanisms Summary

Curly arrow: Represents the movement of a pair of electrons. A curly arrow starts from an electron-rich source (lone pair or bond) and points to an electron-deficient destination. A single-headed (fishhook) arrow represents the movement of a single electron (used in radical mechanisms).

Types of organic reaction:

- **Addition:** Two molecules combine to form one product (atoms add across a double bond).
- **Substitution:** An atom or group is replaced by another atom or group.
- **Elimination:** A small molecule (e.g. H_2O , HBr) is removed from a larger molecule, forming a double bond.
- **Oxidation/Reduction:** Gain or loss of oxygen/hydrogen, or change in oxidation state.
- **Hydrolysis:** A bond is broken by the addition of water (or OH^-).

Exam Tip: When drawing mechanisms, every curly arrow must start from a lone pair or bond and point to where the electrons are going. Show relevant partial charges ($\delta+$ and $\delta-$) on polar bonds. For ionic intermediates, show the charge clearly. Marks are awarded for correct curly arrows, intermediate species, and products.